

Lab 7 Activity

1

Load in the `PlantGrowth` data from the `datasets` package. You can do so by running:

```
data("PlantGrowth", package = "datasets")
```

These are the data discussed in the first part of the one-way ANOVA lecture. The IV is `group` and the DV is `weight`.

2

Test the assumptions of your ANOVA using the methods we learned in earlier weeks to test assumptions about normality and outliers.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.1      v stringr    1.5.2
v ggplot2    4.0.0      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0

-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
## outliers
# from slide 4 in the lecture slides, trt1 might have some outliers
PlantGrowth %>% group_by(group) %>%
  summarize(mean = mean(weight), tmean = mean(weight, trim = .1), .groups = "keep")
```

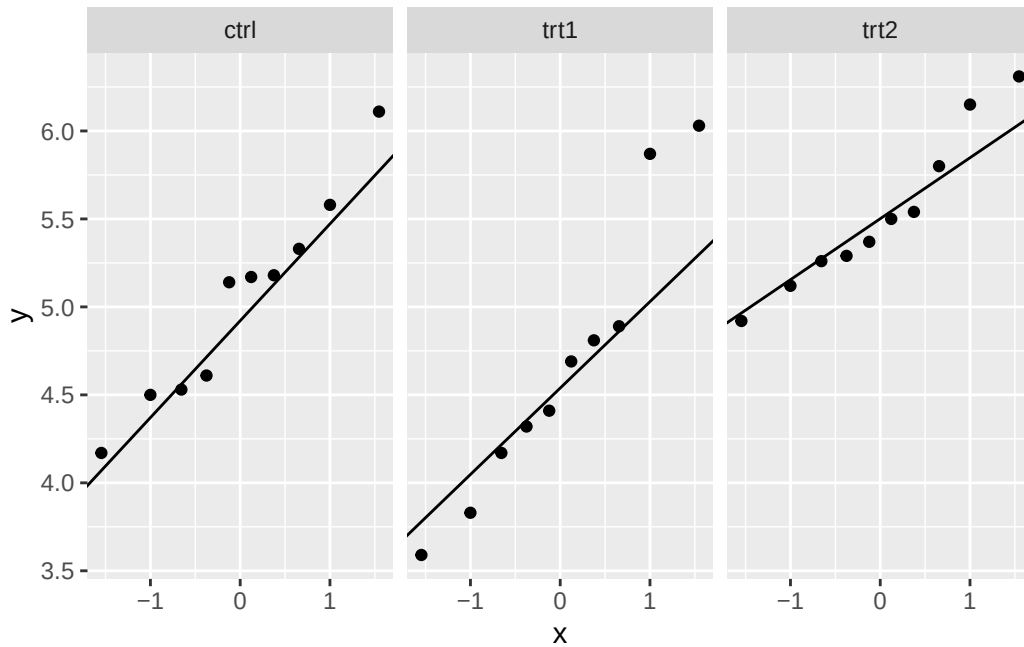
```
# A tibble: 3 x 3
# Groups:   group [3]
  group mean tmean
<fct> <dbl> <dbl>
1 ctrl  5.03  5.00
2 trt1  4.66  4.62
3 trt2  5.53  5.50
```

```
# no highly influential outliers here
```

```
## normality
PlantGrowth %>% group_by(group) %>%
  summarize(skew = e1071::skewness(weight),
            kurt = e1071::kurtosis(weight), .groups = "keep")
```

```
# A tibble: 3 x 3
# Groups:   group [3]
  group skew kurt
<fct> <dbl> <dbl>
1 ctrl  0.231 -1.12
2 trt1  0.474 -1.10
3 trt2  0.485 -1.16
```

```
# potentially significant skew and kurtosis
ggplot(PlantGrowth, aes(sample = weight)) +
  stat_qq() + stat_qq_line() + facet_wrap(~group)
```



```
# possibly some deviations from normality, but
# n is so small that it's hard to tell
# Robust tests are more advisable
```

3

Run an ANOVA that is appropriate for these data using the package of your choice, and compute an effect size.

```
# a number of options available, a few are shown below
```

```
data("PlantGrowth", package = "datasets")
```

```
# aov without weighted least squares
```

```
summary(aov(weight ~ group, PlantGrowth))
```

```

              Df Sum Sq Mean Sq F value Pr(>F)
group          2   3.766    1.8832   4.846 0.0159 *
Residuals     27  10.492    0.3886
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# aov with weighted least squares
```

```
PlantGrowth <- PlantGrowth %>% group_by(group) %>%  
  mutate(weights = 1 / var(weight)) %>% ungroup  
summary(aov(weight ~ group, PlantGrowth, weights = weights))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)  
group          2  10.77    5.383    5.383 0.0108 *  
Residuals     27  27.00    1.000  
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# aov with White's adjustment
```

```
car::Anova(aov(weight ~ group, PlantGrowth), white.adjust = TRUE)
```

Coefficient covariances computed by hccm()

Analysis of Deviance Table (Type II tests)

Response: weight

```
              Df      F    Pr(>F)  
group          2 4.8444 0.01593 *  
Residuals     27
```

```
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ezANOVA - need to add an ID variable
```

```
library(ez)
```

```
PlantGrowth <- PlantGrowth %>% mutate(ID = 1:nrow(PlantGrowth))
```

```
# ezANOVA
```

```
ezANOVA(data = PlantGrowth, dv = weight, wid = ID, between = group) # note that Levene's test
```

Warning: Converting "ID" to factor for ANOVA.

Coefficient covariances computed by hccm()

\$ANOVA

```
  Effect DFn DFd      F      p p<.05      ges  
1 group   2  27 4.846088 0.01590996 * 0.2641483
```

```
$`Levene's Test for Homogeneity of Variance`
  DFn DFd      SSn      SSd      F      p p<.05
1    2   27 0.3495267 4.21611 1.119186 0.3412266
```

```
# oneway with Welch's correction:
library(rstatix)
```

Attaching package: 'rstatix'

The following object is masked from 'package:stats':

filter

```
welch_anova_test(PlantGrowth, weight ~ group)
```

```
# A tibble: 1 x 7
  .y.      n statistic  DFn  DFd      p method
* <chr> <int>    <dbl> <dbl> <dbl> <dbl> <chr>
1 weight    30      5.18     2  17.1 0.017 Welch ANOVA
```

```
library(effectsize)
```

Attaching package: 'effectsize'

The following objects are masked from 'package:rstatix':

cohens_d, eta_squared

```
res <- aov(weight ~ group, PlantGrowth)
```

```
eta_squared(res)
```

For one-way between subjects designs, partial eta squared is equivalent to eta squared. Returning eta squared.

```
# Effect Size for ANOVA
```

```
Parameter | Eta2 |      95% CI  
-----  
group     | 0.26 | [0.04, 1.00]
```

```
- One-sided CIs: upper bound fixed at [1.00].
```

```
omega_squared(res)
```

For one-way between subjects designs, partial omega squared is equivalent to omega squared. Returning omega squared.

```
# Effect Size for ANOVA
```

```
Parameter | Omega2 |      95% CI  
-----  
group     | 0.20 | [0.00, 1.00]
```

```
- One-sided CIs: upper bound fixed at [1.00].
```

4

Identify an *a priori* contrast to test on the `PlantGrowth` data, and test it in R.

```
library(emmeans)
```

Welcome to emmeans.

Caution: You lose important information if you filter this package's results.

See '? untidy'

```
# one option is to compare the control group to the average of the two treatments  
summary(PlantGrowth$group) # factor levels are ordered ctrl, trt1, trt2
```

```
ctrl trt1 trt2  
  10   10   10
```

```
aov_res <- aov(weight ~ group, PlantGrowth)
em1 <- emmeans(aov_res, specs = "group")

contrast(em1, method = list(ctr_v_trt = c(2, -1, -1)))
```

```
contrast estimate SE df t.ratio p.value
ctr_v_trt -0.123 0.483 27 -0.255 0.8009
```

```
# not statistically significant, but bear in mind that n is small
```

5

Conduct post-hoc tests of your choosing on the `PlantGrowth` data. What groups are significantly different from each other with your choice of adjustment for multiple comparisons? In real life, would you recommend conducting these post-hoc tests, and why?

```
# yes, it's appropriate to look at these post-hoc tests since the original ANOVA
# is significant at alpha = .05
```

```
# any package is fine
# most methods show that the major difference is between trt1 and trt2
```

```
em_res <- emmeans(aov_res, specs = "group")
pairs(em_res, adjust = "holm")
```

```
contrast estimate SE df t.ratio p.value
ctrl - trt1 0.371 0.279 27 1.331 0.1944
ctrl - trt2 -0.494 0.279 27 -1.772 0.1754
trt1 - trt2 -0.865 0.279 27 -3.103 0.0134
```

P value adjustment: holm method for 3 tests

```
pairs(em_res, adjust = "bonferroni")
```

```
contrast estimate SE df t.ratio p.value
ctrl - trt1 0.371 0.279 27 1.331 0.5832
ctrl - trt2 -0.494 0.279 27 -1.772 0.2630
trt1 - trt2 -0.865 0.279 27 -3.103 0.0134
```

P value adjustment: bonferroni method for 3 tests

```
pairs(em_res, adjust = "tukey")
```

contrast	estimate	SE	df	t.ratio	p.value
ctrl - trt1	0.371	0.279	27	1.331	0.3909
ctrl - trt2	-0.494	0.279	27	-1.772	0.1980
trt1 - trt2	-0.865	0.279	27	-3.103	0.0120

P value adjustment: tukey method for comparing a family of 3 estimates